

Topic 1.10 - Rational Functions and Holes

Guided Notes

Strengthened ECHS edition - reasoning, challenge, and AP-style practice

Name: _____

Class: _____

Date: _____

Why this topic matters

A hole is not simply a missing dot on a graph; it records an input excluded by the original rule even though a simplified expression has a finite nearby value. This distinction matters when simplifying, graphing, repairing a function, and interpreting equivalent expressions.

Learning targets

- Identify holes from canceled factors and determine exact coordinates.
- Distinguish holes from zeros and vertical asymptotes while preserving the original domain.
- Define repaired piecewise functions and construct rational functions with specified holes.

Language and structure

Term	Meaning in this topic
hole	A removable discontinuity caused by a canceled denominator factor.
removable discontinuity	An excluded input with a finite limiting value.
hole coordinate	$(a, g(a))$, where a is excluded and g is the simplified rule.
continuous repair	Defining the missing output to equal the simplified value.
equivalent on a domain	Two rules agree for all inputs in the stated common domain.

Launch - reason before calculating

Analyze $R(x) = \frac{x^2-9}{x^2-x-6}$. Identify every excluded input and determine which is a hole and which is a vertical asymptote.

Concept 1: Finding a hole and its coordinate

Core idea

Record original restrictions, cancel common factors, then substitute the canceled input into the simplified rule. The resulting finite value is the hole's y -coordinate, not a value of the original function.

Worked example - annotate the reasoning

Problem. Find the hole of $R = \frac{x^2-9}{x^2-x-6}$.

Model reasoning. Simplified $(x+3)/(x+2)$ with $x \neq 3, -2$. At $x = 3$, the simplified value is $6/5$, so the hole is $(3, 6/5)$.

Check your understanding

Independent

No calculator

Find the hole of $F = \frac{(x-1)(x+4)}{(x-1)(x-2)}$.

Concept 2: Holes, zeros, and remaining factors

Core idea

After cancellation, a canceled input can lie on the x -axis of the simplified graph, but it remains a hole rather than a zero. If part of a repeated denominator factor remains after cancellation, the input is still a vertical asymptote, not a hole.

Worked example - annotate the reasoning

Problem. Analyze $G = \frac{(x-1)^2(x+4)}{(x-1)(x-2)}$.

Model reasoning. Simplified $(x-1)(x+4)/(x-2)$. The canceled input 1 creates a hole at $(1, 0)$; it is not a zero of the original function. $x = 2$ is a vertical asymptote.

Check your understanding

Independent

No calculator

Analyze $H = \frac{(x+3)(x-2)}{(x+3)^2}$.

Concept 3: Repair and construction

Core idea

A hole at $x = a$ is repaired by defining the missing value to equal the simplified expression at a . To construct a hole, multiply numerator and denominator by $(x - a)$; choose the simplified rule first so that its value at a is the desired height.

Worked example - annotate the reasoning

Problem. Repair $R = \frac{x^2-9}{x^2-x-6}$ at $x = 3$.

Model reasoning. Define $F(x) = R(x)$ for $x \neq 3, -2$ and $F(3) = 6/5$. The vertical asymptote at -2 cannot be repaired by assigning one finite value.

Check your understanding

Independent

No calculator

Construct a rational function with hole $(2, 5)$, vertical asymptote -1 , and zero 4 .

Representation laboratory

Connect at least two representations

A graph looks like $y = (x + 1)/(x - 3)$ but has open circles at $x = 0$ and $x = 2$. Write one original rational expression that creates exactly those holes and state their coordinates.

Challenge laboratory

Challenge - justify every claim

Construct a rational function with holes at $(1, 2/5)$ and $(2, 1/2)$, vertical asymptote -4 , and no zeros other than the zero of its simplified numerator.

Error clinic and calculator strategy

Common errors to diagnose

- Substituting into the original $0/0$ form to obtain a hole coordinate.
- Calling a canceled input a zero because the simplified numerator also vanishes there.
- Forgetting noncanceled denominator factors after partial cancellation.
- Claiming the original and simplified rules have identical domains.

Technology decision

Graphing software may not display a tiny hole. Compare original and simplified tables, inspect the original domain, and use an open-circle annotation. Exact factoring is more reliable than visual detection.

AP-style synthesis**Multi-part reasoning task**

Let $R(x) = \frac{(x-1)(x+2)^2}{(x-1)(x+2)(x-3)}$.

A. State all original exclusions and simplify.

B. Identify every hole and its coordinate and every vertical asymptote.

C. Define a piecewise repair at all removable discontinuities.

Exit ticket**Before you leave**

1. How is a hole's y -coordinate found?

2. Can a hole lie on the x -axis without being a zero?

3. What value repairs a removable discontinuity?
